High-Precision Satellite Positioning Receiver User Manual

BEIDOUiN

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1. Connecting and Viewing the Board

1.1 Power Supply and Antenna Connection

The Mosaic development board can be powered via a USB cable. It comes with an SMA antenna connector, allowing users to directly connect an antenna.

1.2 Accessing the Web Interface

Once the device is connected to a computer via USB, you can typically access its web interface by entering the address 192.168.3.1 in a browser.

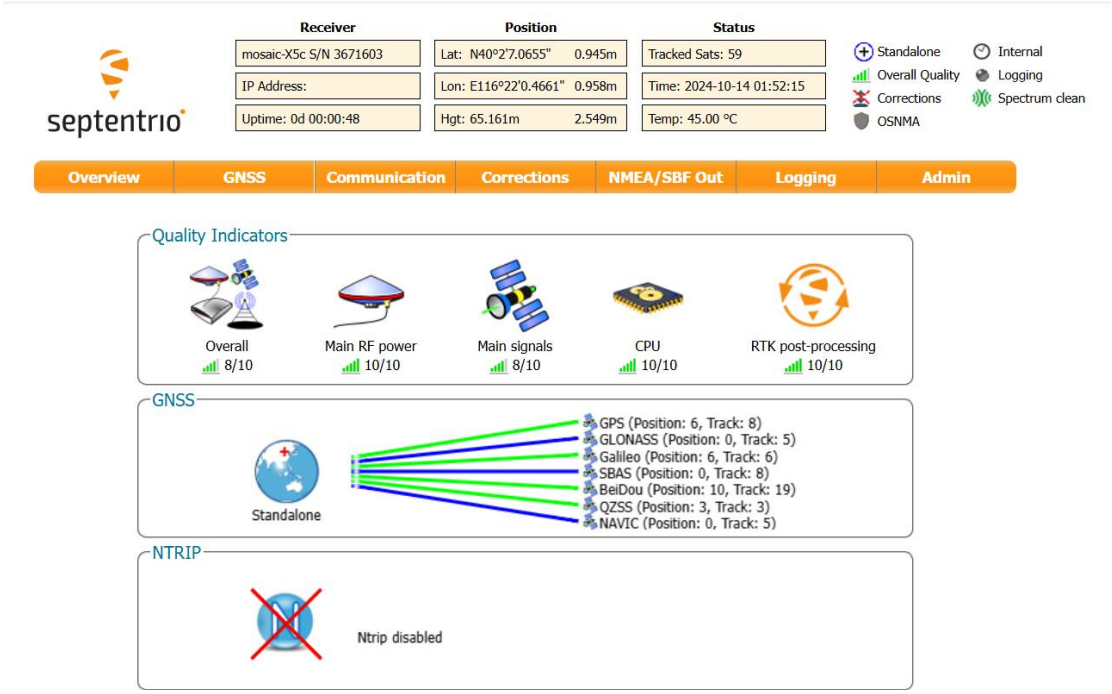


Figure 1: Mosaic Web Interface

If the web interface doesn’t load, it may be due to an unrecognized driver. Check in "Device Manager" to see if the device is recognized, and manually install the driver if needed (the driver can be found in the Mosaic Firmware Package under the driver

folder). You can also access the web interface through a configured IP address, with instructions on configuring the IP provided later.

1.3 Viewing Board Status

1.3.1 Web Interface Status Information

The top of the web interface shows status information, such as the serial number, positioning status, and satellite count.

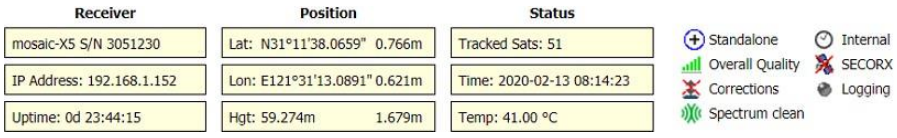
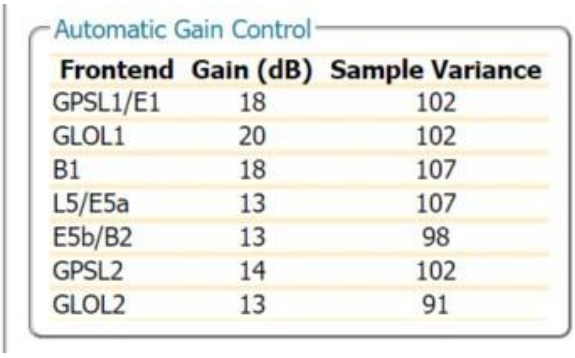


Figure 2: Mosaic Web Interface Top Bar Information

1.3.2 Viewing Antenna Gain

Normal antenna gain ranges between 15-50dB. A higher gain may indicate poor antenna connection or excessive cable loss. You can check this under GNSS -> Spectrum -> Automatic Gain Control.



Automatic Gain Control

Frontend	Gain (dB)	Sample Variance
GPSL1/E1	18	102
GLOL1	20	102
B1	18	107
L5/E5a	13	107
E5b/B2	13	98
GPSL2	14	102
GLOL2	13	91

Figure 3: AGC Table

1.3.3 Viewing Satellite Acquisition and Signal-to-Noise Ratio

Satellite acquisition and signal-to-noise ratio (SNR) can be viewed under GNSS -> Satellites and Signals.

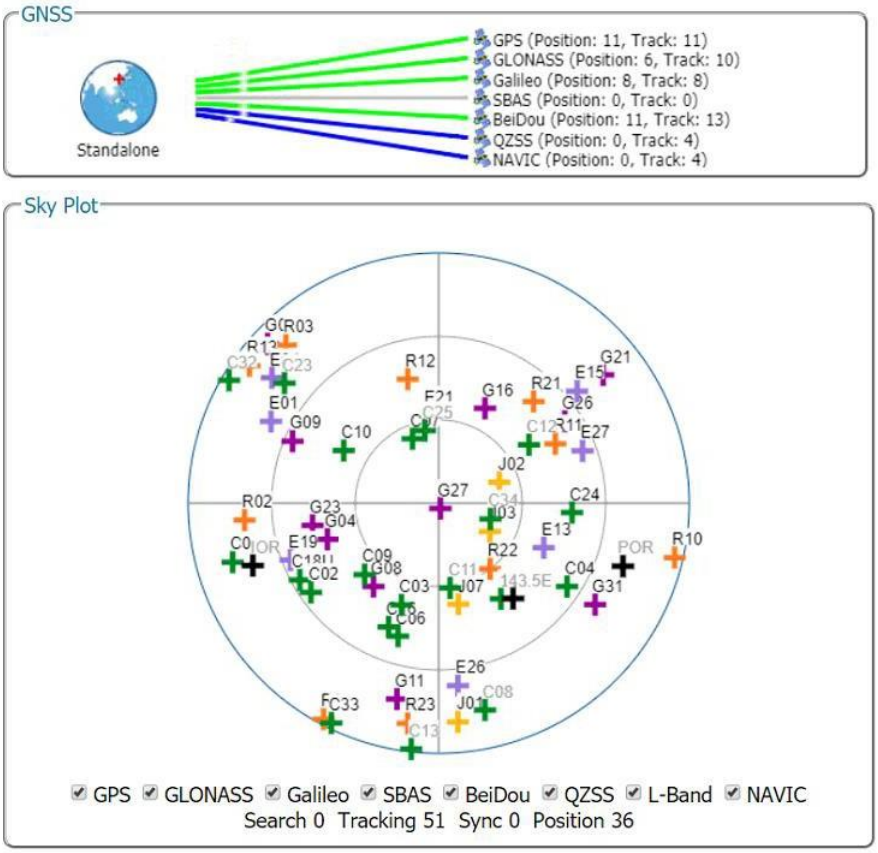


Figure 4: Satellite Acquisition and Visible Satellites Chart

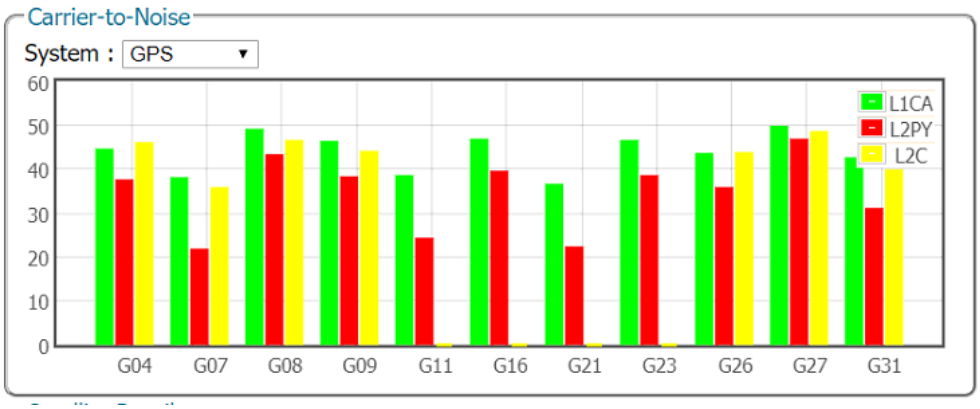


Figure 5: Signal-to-Noise Ratio

1.3.4 Viewing Positioning Status

Positioning status can be found under GNSS -> Position -> Position Information.

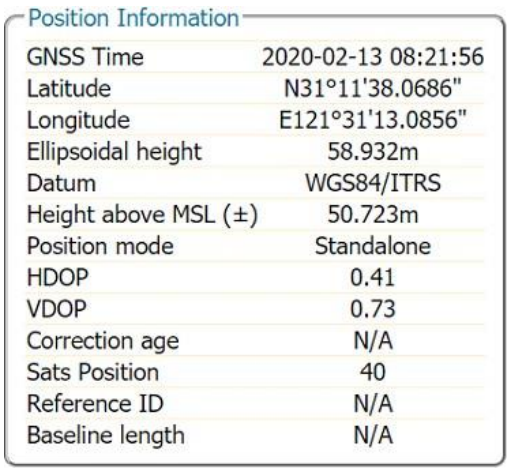


Figure 6: Positioning Status

2. Configuring RTK

2.1 Rover Configuration

In the default configuration, the device automatically detects the data format from each input port and uses differential data accordingly. No special configuration is needed for the rover. To check

if the device is in rover mode, go to GNSS -> Position -> Position Mode to see if it is set to "Rover."

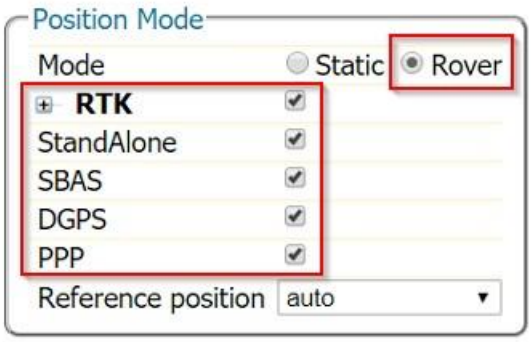


Figure 7: Rover Configuration

Differential data input can be checked under Corrections -> Correction Input. If differential data is received, it will be displayed in the DataStream section.

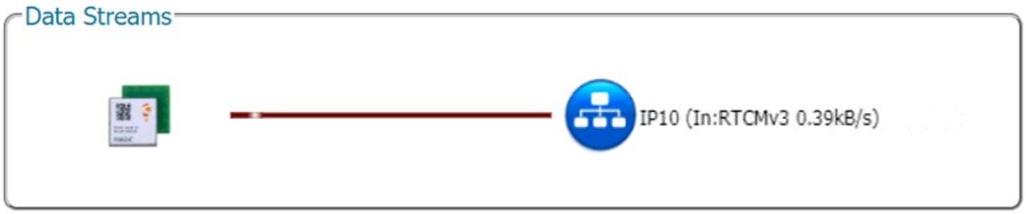


Figure 8: Differential Data Input

2.2 Base Station Configuration

2.2.1 Configuring a Fixed Base Station

Go to GNSS -> Position -> Position Mode to configure the base station.

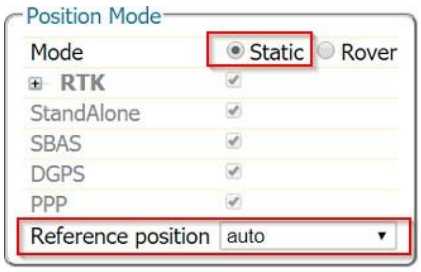


Figure 9: Fixed Base Station Position

Note: If the base station is set to "auto," Mosaic will use the last known valid position as the base station coordinates. For applications requiring precise positions, using "auto" is not recommended. Instead, you can manually enter known point coordinates and select either "Geodetic" or "Cartesian" coordinates in Advanced Settings -> Antenna Reference Point Static Position.

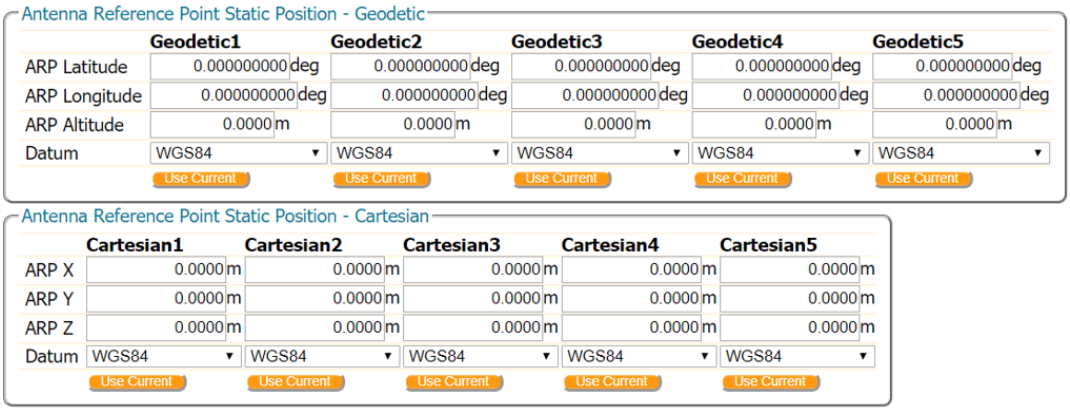


Figure 10: Set Fixed Base Station Coordinates

After configuration, click “OK” at the bottom of the page to apply the settings.

2.2.2 Configuring Differential Data Output

1) Differential data output can be configured under Corrections -> Corrections Output. For example, to output RTCMv3 data from the COM1 port:

2) Click New RTCM3 Output under the Differential Corrections Output section.

3) Select Serial Port, then click Next.

4) Choose COM1, and click Next.

5) 选Select the differential data types to output, including GPS, GLONASS, Beidou, and Galileo.

6) After selection, click Finish.

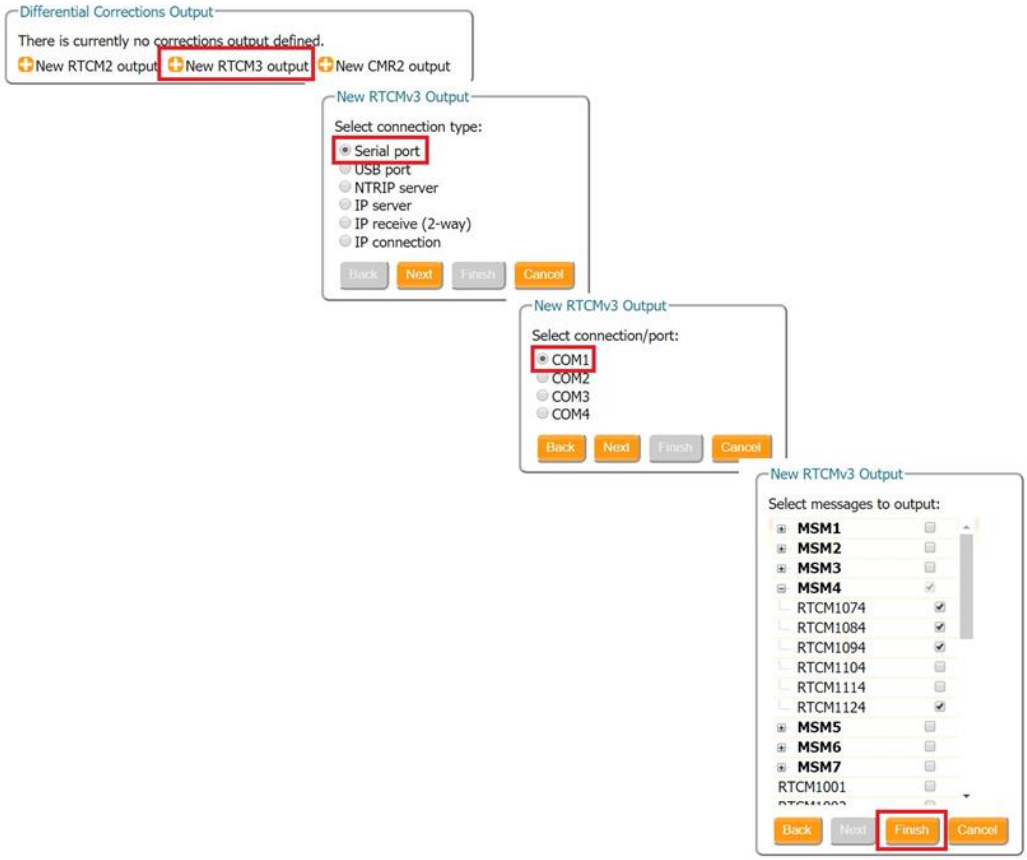


Figure 11: Configure COM1 Port for RTCMv3 Output

After all configurations, click OK to apply the settings.

3. Configuring NMEA/SBF Data Output

NMEA and SBF data output can be configured under NMEA/SBF Output in the web interface. Below is an example of configuring COM1 to output 1Hz NMEA GGA+GSV data:

1) Select New NMEA Stream under NMEA/SBF Output.

2) Choose Serial Port, then click Next.

3) Select COM1, then click Next.

4) Select the output frequency, check GGA+GSV, and click Finish.

5) More NMEA settings can be found under Advanced Settings.

6) Click OK at the bottom to apply the settings.

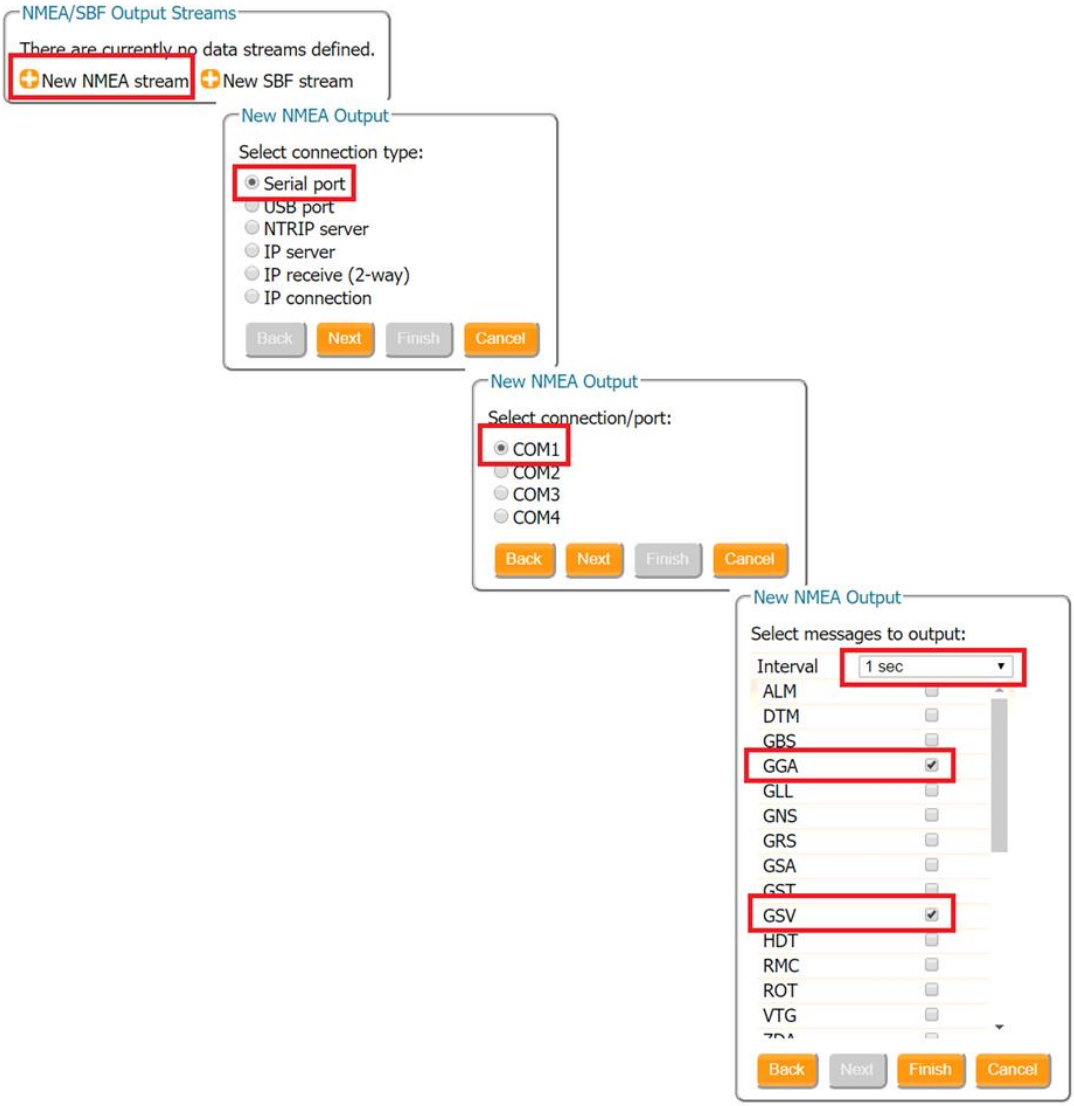


Figure 12: Configure COM1 Port for NMEA GGA+GSV Data Output

4. Network Configuration

4.1 Configuring Network Communication

4.1.1 Configuring IP Address

The IP address can be configured via Communication -> Ethernet in the web interface:

1) Enable the Ethernet Interface Mode and set Power to on.

2) Configure the IP address under TCP/IP Settings. If using DHCP mode, Mosaic will automatically obtain an IP address from the router. For Static mode, manually input the IP address, gateway, and other parameters.

3) After configuration, click OK to apply the settings.

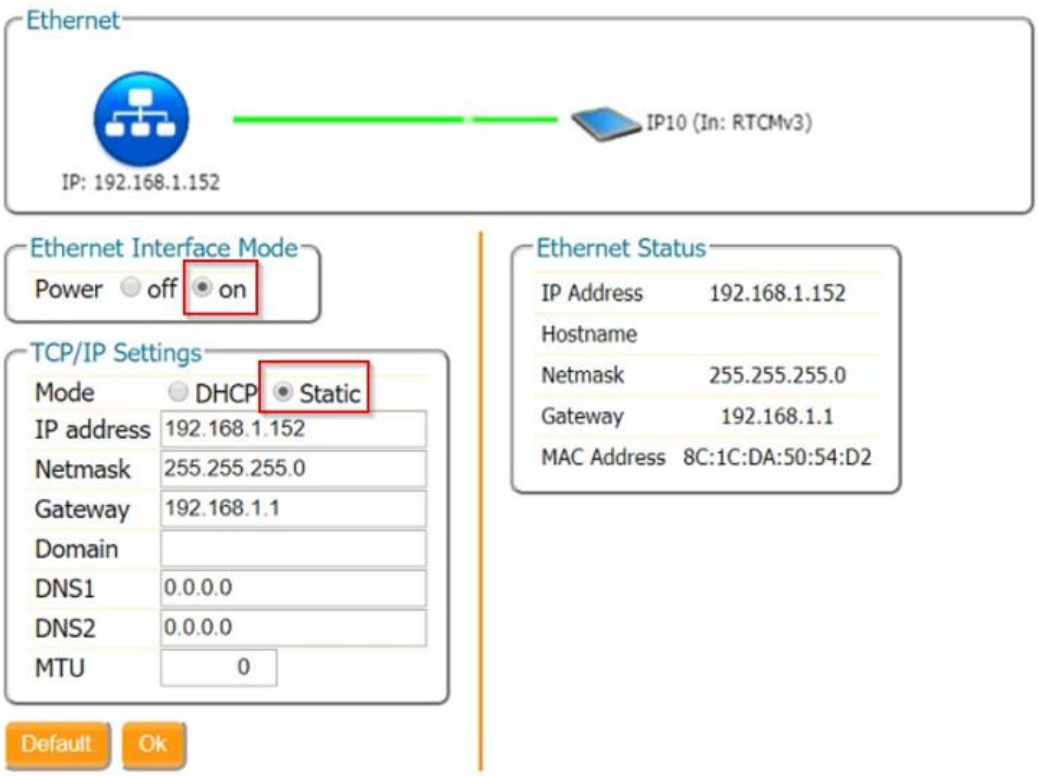


Figure 13: Set IP Address

4.1.2 Configuring IPS/IPR Connection

IP Server (IPS) or IP Receive (IPR) connections can be set up under Communications -> IP Ports. To establish a new connection, click New IP Server/New IP Receive Connection, set the communication method, port number, and other parameters, then click OK to apply the settings.

4.1.3 Saving NMEA Data

To save NMEA data, follow the same steps as for saving SBF data.

Navigate to RxControl -> Communication -> Output Settings -> NMEA Output… to configure it.

4.1.4 Changing SD Card Logging and Naming

In RxControl, you can adjust the settings for managing SD card storage space and file naming via Logging -> Internal Logging Settings….

4.1.5 Mounting/Unmounting/Formatting SD Card and Downloading Data

To mount/unmount/format the SD card, go to RxControl -> Logging -> Disk Management. After unmounting the SD card, it can be accessed directly from a PC via USB.

Note: Before removing the SD card, please stop data recording first (refer to section 5.1.1, where the update rate can be set to "off"), otherwise it may result in data loss.

4.2 Configuring NTRIP

4.1.1 Configuring NTRIP Client

You can add a new NTRIP client connection by going to Corrections -> NTRIP and clicking New NTRIP Client. Fill in the required information, including the server address, port, username, password, and mount point. After filling in the details, click OK to apply the settings.

The image below shows an example of a configuration for connecting to the Qianxun NTRIP service.

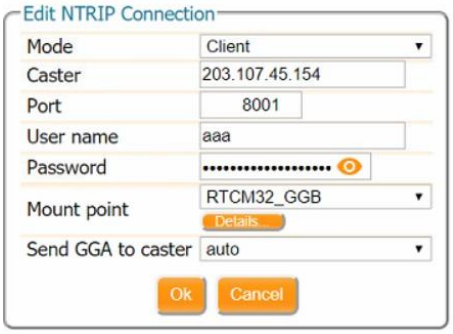


Figure 14: NTRIP Client Configuration

After configuration, the NTRIP status bar at the top will display the connection status. If successfully connected, it will turn green, showing the connected server address, mount point, and data format.

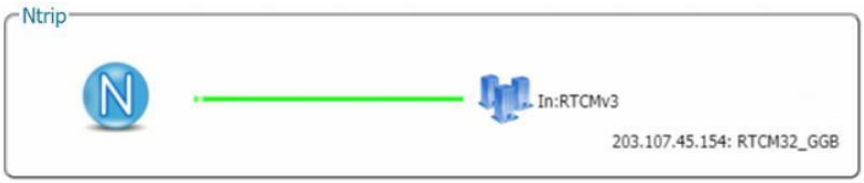


Figure 15: NTRIP Client Status

4.1.2 Configuring NTRIP Server and Caster

To set up an NTRIP server, go to Corrections -> NTRIP and click New NTRIP Server.

When configuring the NTRIP server, you can enable the internal caster by selecting Internal Caster. The settings for the internal NTRIP caster can be found under Communication -> NTRIP Caster:

1) Enable the NTRIP Caster under General Settings, and set the port number.

2) Click New Mount Points to configure new mount points. The image below shows an example configuration of the Mosaic_1 mount point, outputting RTCMv3 data.

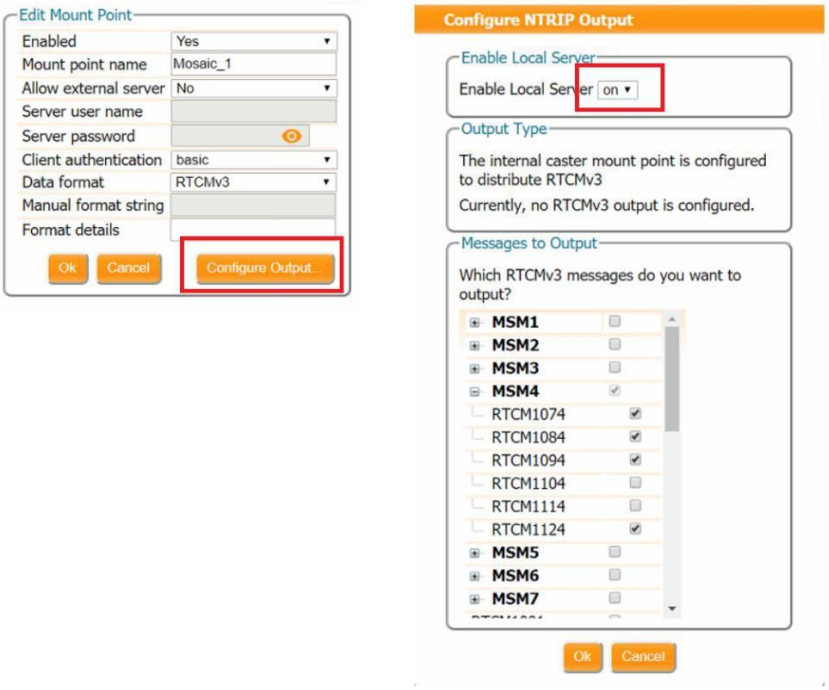


Figure 16: Configuring Mount Points

3) In the Client Users section, define the username and password for connecting to the caster.

4) Click OK at the bottom to apply the settings.

If you want to send data to an external caster, you will need to enter the external caster's address, port number, and other details in the IP Server settings interface.

5. Advanced Control Panel

The Advanced Control Panel (located in Admin -> Expert Control -> Control Panel) allows users to modify more detailed module settings, including satellite/signal tracking and elevation masks. When making changes in the advanced settings, ensure that you fully understand the purpose and impact of each modification.

The following sections explain some common settings in the advanced panel.

5.1 Configuring Elevation Masks

Elevation masks can be adjusted under Navigation -> Receiver Operation -> Masks. Changing the Tracking elevation mask will stop tracking satellites below the specified angle, while changing the PVT (Position, Velocity, Time) mask will stop using satellites below the specified angle for calculations. If the PVT angle is lower than the Tracking angle (e.g., Tracking = 15°, PVT = 10°), only satellites above 15° will be tracked and used for calculations.

5.2 Configuring Satellite and Signal Tracking

Satellite tracking settings can be modified under Navigation -> Receiver Operation -> Advanced User Settings -> Satellite Tracking, where you can enable or disable tracking for specific satellite systems or individual satellites.

Signal tracking can be adjusted under Navigation -> Receiver Operation -> Advanced User Settings -> Signal Tracking, allowing you to disable tracking for certain satellite systems or signals.

Note: Disabling certain satellite systems or signal tracking may impact Mosaic’s ability to acquire time or ephemeris data, which could affect all satellite tracking. Before disabling a satellite system, confirm that it is not required for time acquisition (check GNSS -> PPS/Timing).

6. Management and Firmware Upgrades

6.1 Saving Configuration

Mosaic does not save configurations by default, so after a power cycle, it will revert to its default settings. If you need to save your current configuration to the startup settings, go to Admin -> Configurations. After making changes in other interfaces, a dialog

box will appear in the lower right corner asking if you want to save the current configuration to the startup settings. You can also save the configuration by clicking on this dialog box.

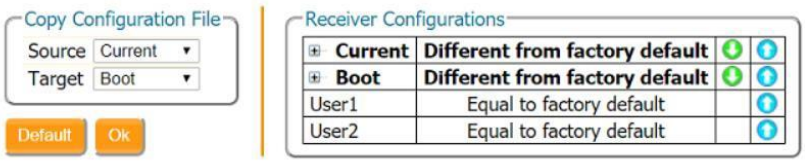


Figure 17: Save Current Configuration to Startup Settings

On the right side of the screen, you can download the configuration as a text file by clicking the green button or upload a configuration file by clicking the blue button.

6.2 User Management

User management can be accessed through Admin -> User Administration, where you can add users and assign different permission levels.

Click New User to add a user, set the username, password, and access permissions. In the Default Access Level Per Interface section, you can configure whether each interface requires login and if access to Mosaic is permitted. Setting permissions to none means no information can be viewed without logging in, while Viewer permissions allow viewing but not modifying settings.

Note: For security reasons, once a username/password is set, it cannot be recovered or reset through regular means. If you forget your password, you will need to contact the manufacturer for recovery.

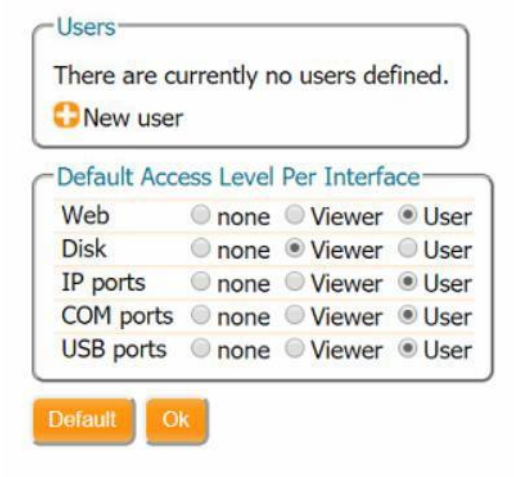


Figure 18: User Management Interface

6.3 Firmware Upgrade

Firmware upgrades can be performed under Admin -> Upgrade. Click Choose File to select the local firmware file, then click Start Upgrade to upload the firmware. The module will automatically restart during the upgrade. For the correct upgrade sequence, refer to the Release Notes in the firmware package.

6.4 Resetting the Module

You can reset the module by going to Admin -> Reset.

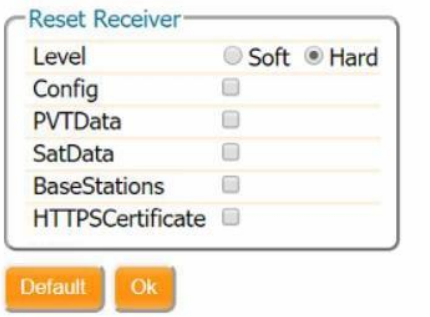


Figure 19: Reset Interface

There are two types of reset methods: Soft reset only resets the firmware, while Hard reset also resets the hardware. When no other options are checked, a Hard Reset is equivalent to restarting the module.

In the options below, Config represents erasing all configurations of the board, while PVTData, SatData, and BaseStations represent erasing the position data, ephemeris data, and base station data stored in NVRAM, respectively.

6.5 Viewing Firmware Version, Serial Number, and Other Information

Module information, including the serial number and firmware version, can be found under Admin -> About.

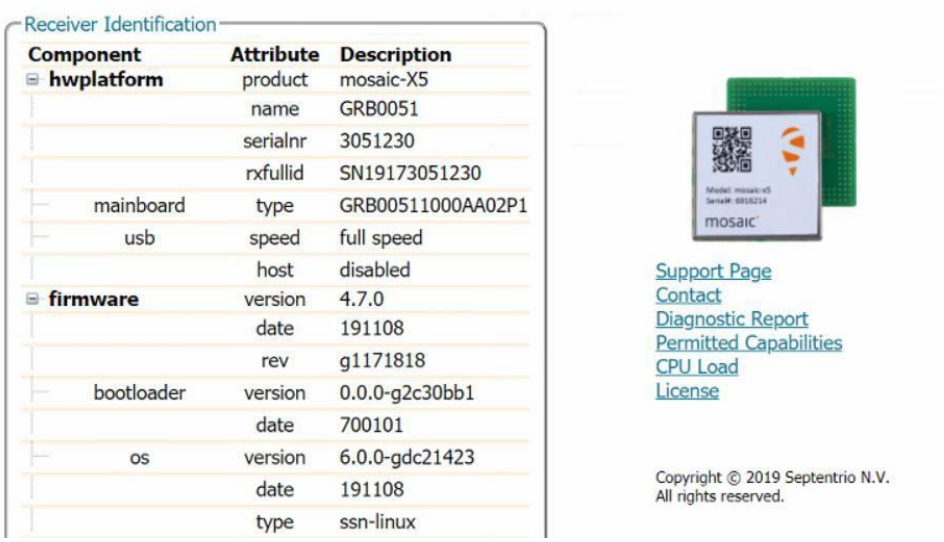


Figure 19: Mosaic Information Interface

Click Diagnostic Report on the right to generate a report. This report contains important technical information and can be sent to technical support when needed.